

AuRoRA-2W

Autonomous Road-scene Recognition Architecture for 2-Wheelers

Modular Phased Implementation Plan

Purpose of This Document

This is the modular build plan for the AuRoRA-2W codebase, updated to utilize the MOTOR dataset (arXiv 2026). Each phase is designed to be highly independent, possessing a concrete goal, specific actionable steps, and a definitive validation check. This ensures that any AI tool assisting with a specific phase requires minimal prior context to implement its portion correctly.

MOTOR Dataset Context

The MOTOR dataset provides genuine video synced with raw IMU telemetry at 30Hz. However, it provides angular velocity (gyro), not absolute roll angle. The dataset is organized into maneuver-specific clips.

Phase 0: Dataset Acquisition & Verification

Goal: Obtain a synchronized video and raw telemetry pair containing a high-lean maneuver, directly from the MOTOR dataset.

Context for AI Agent: This phase focuses on manual/targeted data retrieval without bulk scraping. The telemetry is already perfectly time-aligned to the video frames (30Hz).

Actions:

- Access the `varunpaturkar/MOTOR` repository on Hugging Face.
- Download the root `annotations.csv` file to identify specific maneuver clips (target tags: `left_turn`, `right_turn`, `lane_change`, or `overtake`).
- Navigate the `clips/` directory structure to manually download one matched pair: the front-facing video (`front/<clip>.mp4`) and its corresponding telemetry (`telemetry/<clip>.csv`).

Validation:

- Inspect the downloaded CSV to confirm the schema contains `frame`, `timestamp_s`, and `gyro_x/y/z` columns.
- Verify the telemetry is resampled at 30 Hz (matching typical video framerate).

Phase 1: Gyroscope to Roll-Angle Conversion

Goal: Process the raw angular velocity data to generate a clean, drift-corrected, per-frame absolute roll angle (in degrees).

Context for AI Agent: The input data is time-aligned angular rate (gyro). The output must be absolute orientation to drive downstream geometric offset modules. A Kalman filter is strictly required to handle drift.

Actions:

- Ingest the telemetry CSV and isolate the specific gyroscope axis representing the vehicle's roll.
- Integrate the angular velocity over time to compute the raw roll angle.
- **Implement a Kalman filter** to fuse the gyro data (and accelerometer data if available in the schema) to correct for integration drift and stabilize the continuous roll angle curve.
- Output a processed CSV containing: frame index, timestamp, and the final filtered absolute roll angle (degrees).

Validation:

- Write a script using OpenCV to overlay the computed roll value as on-screen text onto the video frames.
- Visually scrub through the composite video: the numerical roll value must noticeably swing during the physical turns and return near zero on straightaways.

Phase 2: Baseline PIDNet Setup (End-to-End)

Goal: Establish a working, unmodified segmentation baseline on the downloaded MOTOR footage.

Context for AI Agent: Ensure the base network architecture functions correctly before introducing any structural modifications. No new model logic is needed here.

Actions:

- Clone the reference PIDNet architecture (e.g., XuJiacong/PIDNet) into a clean `models/` directory. Do not modify this baseline.
- Acquire pre-trained weights (Cityscapes/IDD) or run a brief training loop (a few epochs) to achieve a functional forward pass.
- Run inference on extracted frames from the MOTOR clip acquired in Phase 0.

Validation:

- The segmentation output must roughly outline the drivable road area as a contiguous region. High accuracy is not required, but structural sanity (road is one blob, sky is ignored) is essential.

Phase 3: IDFA Module (Geometric Roll Compensation)

Goal: Implement the core Image Deformation Field Attention (IDFA) module and integrate it via deformable convolutions.

Context for AI Agent: This is a standalone mathematical and structural modification. It takes an absolute roll angle and outputs a spatial offset field.

Actions:

- Implement a closed-form rotation offset computation: for a given roll angle θ and image dimensions, calculate per-pixel $(\Delta x, \Delta y)$ offsets that counteract rotation around the image center.
- Wrap this logic into an independent `idfa.py` module.
- Add 1-2 learnable convolutional layers on top of the closed-form offset for minor geometric refinements (pitch, lens distortion).
- Create `aurora2w.py` (copying the baseline PIDNet) and replace specific standard convolutions in the Proportional branch with deformable convolutions that accept the IDFA offset field.

Validation:

- **Unit Test:** Feed `idfa.py` a synthetic roll angle (e.g., 20°) and a test image grid. Visually confirm the offset field shifts sample points in the correct geometric direction.
- Run a forward pass of `aurora2w.py` on a zero-roll frame and a high-roll frame without shape errors.

Phase 4: Qualitative Comparison

Goal: Validate the efficacy of the IDFA module on actual high-roll footage.

Context for AI Agent: Compare the outputs of the baseline model against the modified model using the processed data.

Actions:

- Using the processed CSV from Phase 1, identify frames with the highest absolute roll angles.
- Run both the baseline PIDNet (Phase 2) and AuRoRA-2W (Phase 3) on these selected frames.
- Generate side-by-side visual segmentation masks for comparison.

Validation:

- Observe a qualitative improvement: at high roll, AuRoRA-2W's road segmentation should appear more stable and contiguous compared to the baseline, which may exhibit distortion or broken boundaries.

Phase	Core Deliverable	Independent Validation Check
0: Data	MOTOR clip (.mp4) + Telemetry (.csv)	CSV schema verified; 30Hz resampled data confirmed.
1: Filter	Processed CSV (absolute roll angle)	OpenCV video overlay: Roll values swing accurately during turns.
2: Baseline	Working baseline PIDNet inference	Output segmentation correctly identifies road as a sane, contiguous blob.
3: IDFA	idfa.py + aurora2w.py modules	Offset field shifts correctly on synthetic grid; forward pass executes without shape errors.
4: Compare	Side-by-side segmentation visual	

Phase	Core Deliverable	Independent Validation Check
		AuRoRA-2W shows visibly more stable contiguous boundaries at high lean angles vs baseline.